

dilute gas

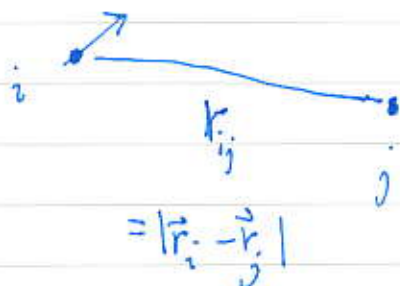
$$H = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m} + \sum_{i < j} V_{ij}$$

$\sum_{i < j} V_{ij}$ pair potential
 $\uparrow$ interaction

eg. Lennard-Jones potential

Date

$$\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6$$



$$V_{ij} = V(r_{ij})$$

$$T \propto e^{-\beta(H - \mu N)}$$

$$Z = \sum_{N=0}^{\infty} \frac{1}{N!} \int \frac{d\vec{r}_N}{h^{3N}} e^{-\beta(H_N - \mu N)}$$

$$= \sum_{N=0}^{\infty} e^{\beta \mu N} Z_N$$

$$Z_N = \frac{1}{N! h^{3N}} \int d\vec{r}_N e^{-\beta H_N}$$

$$= \frac{1}{N! h^{3N}} \int d\vec{p}_1 d\vec{p}_2 \dots d\vec{p}_N e^{-\beta \sum_{i=1}^N \frac{\vec{p}_i^2}{2m}} \int d\vec{q}_1 d\vec{q}_2 \dots d\vec{q}_N e^{-\beta V_N}$$

$$= \frac{1}{N! h^{3N}} \left[\int_{-\infty}^{\infty} d\vec{p} e^{-\beta \frac{\vec{p}^2}{2m}} \right]^{3N} \underbrace{\int d\vec{r}_1 d\vec{r}_2 \dots d\vec{r}_N e^{-\beta V_N}}_{Q_N}$$

$$= \frac{1}{N! h^{3N}} \left(\sqrt{2\pi} \frac{m}{\beta} \right)^{3N} Q_N$$

configuration partition function

$$= \frac{1}{N!} \left(\frac{\sqrt{2\pi m k_B T}}{h} \right)^{3N} Q_N$$

$$= \frac{1}{N!} \lambda^{3N} Q_N$$

$$\lambda = \frac{h}{\sqrt{2\pi m k_B T}}$$

$$Z = \sum_{N=0}^{\infty} \frac{1}{N!} \left(\frac{e^{\beta \mu}}{\lambda^3} \right)^N Q_N$$

$z = e^{\beta \mu}$ fugacity

$$Q_0 \equiv 1$$

$$Q_1 = \int d\vec{r}_1 e^{-\beta \sum_{i=1}^1 V_i} = V$$

$$Q_2 = \int d\vec{r}_1 d\vec{r}_2 e^{-\beta V(|\vec{r}_1 - \vec{r}_2|)}$$

$$Q_3 = \dots$$

$$\Xi = 1 + \left(\frac{e^{\beta\mu}}{\lambda^3}\right)V + \frac{1}{2}\left(\frac{e^{\beta\mu}}{\lambda^3}\right)^2 V \int_{\text{space}} e^{-\beta V(r)} d\mathbf{r} + \dots \quad 319$$

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$$PV = k_B T \ln \Xi = k_B T$$

$$\Psi = -k_B T \ln \Xi, \quad -N = \frac{\partial \Psi}{\partial \mu} = \frac{e^{\beta\mu}}{\lambda^3}$$

$$= -\frac{1}{\beta} \ln \Xi$$

$$= -\frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu}$$

$$= -\frac{1}{\beta \Xi} \left[\frac{e^{\beta\mu}}{\lambda^3} V + \left(\frac{e^{\beta\mu}}{\lambda^3}\right)^2 V b + \dots \right]$$

$$N = \frac{1}{\Xi} \left[\frac{e^{\beta\mu}}{\lambda^3} V + \left(\frac{e^{\beta\mu}}{\lambda^3}\right)^2 V b + \dots \right] \quad \alpha = \frac{e^{\beta\mu}}{\lambda^3}$$

$$\Xi = 1 + \alpha V + \frac{\alpha^2}{2} b + \dots$$

$$\rho = \frac{N}{V}$$

$$\ln \Xi = \ln \left(1 + \alpha V + \frac{\alpha^2}{2} b + \dots \right)$$

$$\ln(1+x)$$

$$= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$$

$$= \alpha V + \frac{\alpha^2}{2} b - \frac{1}{2} (\alpha V + \frac{\alpha^2}{2} V b)^2 + \dots$$

$$= \alpha V + \frac{\alpha^2}{2} V b - \frac{1}{2} \alpha^2 V^2 + \dots$$

$$PV = k_B T \ln \Xi = k_B T \left[\alpha V + \frac{\alpha^2}{2} V b - \frac{1}{2} \alpha^2 V^2 + \dots \right]$$

$$N = k_B T \frac{\partial \ln \Xi}{\partial \mu} = k_B T \left[\beta \alpha V + \beta (\alpha^2 V b - \alpha^2 V^2) + \dots \right]$$

$$= \alpha V + (\alpha^2 V b - \alpha^2 V^2) + \dots$$

$$PV = k_B T \left[N - \frac{1}{2} (\alpha^2 V b) + \frac{1}{2} (\alpha^2 V^2) + \dots \right] \quad \rho = \frac{N}{V}$$

$$P = k_B T \left(\rho - \frac{1}{2} \alpha^2 \rho \left(\frac{b}{V} \right) + \frac{1}{2} \alpha^2 \rho \right)$$

virial expansion
cluster
Ursell & Mayer

$$\text{Let } e^{-\beta V(r)} = 1 + f(r)$$

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$$Q_N = \int d\vec{r}_1 \dots d\vec{r}_N \prod_{i < j} (1 + f_{ij})$$

$$Q_0 = 1$$

$$Q_1 = V$$

f is small

$$Q_2 = \int d\vec{r}_1 d\vec{r}_2 (1 + f_{12}) = V^2 + V \int 4\pi r^2 dr f(r)$$

$$= \text{two separate circles} + \text{two circles connected by a line}$$

$$Q_3 = \int d\vec{r}_1 d\vec{r}_2 d\vec{r}_3 (1 + f_{12})(1 + f_{13})(1 + f_{23})$$

$$= \text{three separate circles} + \text{two circles connected by a line} + \text{two circles connected by a line} + \text{two circles connected by a line} + \text{three circles connected by lines} + \text{three circles connected by lines} + \text{three circles connected by lines}$$

$$= V^3 + 3V^2 \int 4\pi r^2 dr f(r) + 3V \int d\vec{r}_2 d\vec{r}_3 f_{12} f_{13} \int 4\pi r^2 dr f^2$$

$$\text{let } x = \frac{e^{\beta \mu}}{\lambda^3}$$

$$+ \int d\vec{r}_1 d\vec{r}_2 d\vec{r}_3 f_{12} f_{13} f_{23}$$

$$Z = 1 + \left(\frac{e^{\beta \mu}}{\lambda^3}\right) \cdot V$$

$$+ \frac{1}{2!} \left(\frac{e^{\beta \mu}}{\lambda^3}\right)^2 \left[V^2 + V \int 4\pi r^2 dr f(r) \right]$$

$$+ \frac{1}{3!} \left(\frac{e^{\beta \mu}}{\lambda^3}\right)^3 \left[V^3 + 3V^2 \int 4\pi r^2 dr f(r) + 3V \left[\int 4\pi r^2 dr f^2 \right]^2 + \int d\vec{r}_1 d\vec{r}_2 d\vec{r}_3 f_{12} f_{13} f_{23} \right] + \dots$$

all graphs

$$\Xi = 1 + x [0] + \frac{1}{2!} x^2 [00 + \infty]$$

$$+ \frac{1}{3!} x^3 [000 + \infty + \text{graphs}] + \frac{1}{4!} x^4 [\dots] + \dots$$

all connected graphs

$$\ln \Xi = x [0] + \frac{1}{2!} x^2 [00] + \frac{1}{3!} x^3 [\text{connected graphs}] + \dots$$

check: $\Xi = 1 + xV + \frac{1}{2} x^2 (V^2 + V \int 4\pi r^2 dr f(r))$

$$+ \frac{1}{3!} x^3 (V^3 + 3V^2 \int 4\pi r^2 dr f(r) + 3V [\int 4\pi r^2 dr f]^2 + \int d\vec{r}_1 d\vec{r}_2 d\vec{r}_3 f(\vec{r}_{12}) f(\vec{r}_{13}) f(\vec{r}_{23}) + \dots)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$$

$$\ln \Xi = xV + \frac{1}{2} x^2 (V^2 + Vb) + \frac{1}{6} x^3 (V^3 + 3V^2 b + 3Vb^2 + c) + \dots$$

$$- \frac{1}{2} \left(xV + \frac{1}{2} x^2 (V^2 + Vb) + O(x^3) \right)^2 + \frac{1}{3} (xV)^3 + \dots$$

$$= xV + \frac{1}{2} x^2 Vb + \left(\frac{1}{6} - \frac{1}{2} + \frac{1}{3} \right) x^3 V^3 + \left(\frac{1}{6} \cdot 3 - \frac{1}{2} \right) x^3 V^2 b + \frac{1}{6} x^3 (3Vb^2 + c) + \dots$$

$$P = \frac{k_B T}{V} \ln \Xi = k_B T \left[x + \frac{x^2}{2!} \frac{1}{V} Vb + \frac{x^3}{3!} \frac{1}{V} (3Vb^2 + c) + \dots \right]$$

$$= k_B T \sum_{l=1}^{\infty} x^l b_l$$

$$\underbrace{\quad}_{\frac{1}{V} \ln \Xi}$$

$$b_1 = 1$$

$$b_2 = \frac{b}{2}$$

$$b_3 = \frac{1}{3!} (3b^2 + c)$$

$$b_4 = \dots$$

particle density $\rho = \frac{N}{V} = \frac{1}{V} \frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu} = \frac{1}{V} \frac{\partial \ln \Xi}{\partial (\beta \mu)}$

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$$= \frac{\partial}{\partial (\beta \mu)} \left[\sum_{l=1}^{\infty} x^l b_l \right]$$

$$x = \frac{e^{\beta \mu}}{\lambda^3}$$

$$\rho = \frac{1}{V} \sum_{l=1}^{\infty} l x^l b_l$$

$$\frac{\partial x^l}{\partial (\beta \mu)} = l x^{l-1} \frac{\partial x}{\partial (\beta \mu)}$$

$$= l x^{l-1} \frac{e^{\beta \mu}}{\lambda^3}$$

$$\rho = k_B T \sum_{l=1}^{\infty} x^l b_l = k_B T \sum_{l=1}^{\infty} a_l(T) \rho^l$$

$$= l \frac{x^l}{\lambda^3}$$

$$\rho = x b_1 + 2 x^2 b_2 + 3 x^3 b_3 + \dots$$

virial expansion

$$= x + x^2 b + \frac{1}{2} x^3 (3b^2 + c) + \dots$$

$$x = \rho - x^2 b - \frac{1}{2} x^3 (3b^2 + c) + O(x^4) + \dots$$

$$= \rho - (\rho - x^2 b - \frac{x^3}{2} (3b^2 + c))^2 b - \frac{1}{2} (\rho - x^2 b + \dots)^3 (3b^2 + c) + \dots$$

$$= \rho - \rho^2 b + 2 \rho^3 b^2 - \frac{1}{2} \rho^3 (3b^2 + c) + \dots$$

$$= \rho - b \rho^2 + \frac{1}{2} (b^2 - c) \rho^3 + \dots$$

$$\rho = k_B T \left(x + x^2 \frac{b}{2} + x^3 \frac{1}{6} (3b^2 + c) + \dots \right)$$

$$= k_B T \left(\rho - \rho^2 b + \rho^3 \left(\cancel{2b^2} + \frac{1}{2} b^2 - \frac{c}{2} \right) \right)$$

$$+ \frac{b}{2} (\rho^2 - 2b \rho^3) + \rho^3 \frac{1}{6} (3b^2 + c) + O(\rho^4)$$

$$= k_B T \left[\rho - \frac{b}{2} \rho^2 + \left(\underbrace{2 - \frac{3}{2} - 1 + \frac{1}{2}}_0 \right) b^2 + \left(\frac{1}{6} c \right) \rho^3 + \dots \right]$$

$$a_1 = 1$$

$$= k_B T \left[\rho - \dots \right]$$

$$a_2 = -\frac{b}{2} \leftarrow \text{second virial coefficient}$$

$$= k_B T \left[\rho - \frac{b}{2} \rho^2 + \frac{1}{6} \rho^3 + \dots \right]$$

$$a_3 = -\frac{1}{3} c$$

$$= k_B T \left(\rho - \frac{b}{2} \rho^2 - \frac{c}{3} \rho^3 + \dots \right)$$

graph include the inter

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$$a_n = -\frac{1}{n(n-2)!} \frac{1}{V} \left[\int d\vec{r}_1 \dots d\vec{r}_n \right] \text{ all different } n\text{-point doubly connected graphs}$$

$$a_1 = 1$$

$$a_2 = -\frac{1}{2} \frac{1}{V} \left[\text{---} \text{---} \right]$$

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0.5.2019

$$a_3 = -\frac{1}{3} \frac{1}{V} \left[\text{triangle} \right]$$

$$a_4 = -\frac{1}{4} \cdot \frac{1}{2} \frac{1}{V} \left[3 \text{ square} + 6 \text{ square with diagonal} + \text{square with both diagonals} \right]$$

labeling factor

$$0 = \int d\vec{r}_1 \cdot 1 = V$$

$$\text{---} \text{---} \rightarrow \int d\vec{r}_1 d\vec{r}_2 \cdot f(|\vec{r}_1 - \vec{r}_2|)$$

$$f_{ij} = f_{ji}$$

$$\text{square} = \int d\vec{r}_1 d\vec{r}_2 d\vec{r}_3 d\vec{r}_4 f_{12} f_{23} f_{34} f_{41}$$

R4 Reichl "A modern course in statistical physics"



fix T



5 mark each

- 1 (a) { triple point of water
lowest possible

$$T_t = 273.16 \text{ K} \quad (4)$$

$$T = 0 \quad (1)$$

10 March 08
Date
if 273.15 -1

(b) $U(S, V, N) \quad U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$

$$dU = TdS - pdV + \mu dN$$

$$\rightarrow U = TS - pV + \mu N$$

(c) $\dot{A} = (A, H)$ classical mechanics is possible bracket

$$(A, B) = \sum_j \left(\frac{\partial A}{\partial q_j} \frac{\partial B}{\partial p_j} - \frac{\partial B}{\partial q_j} \frac{\partial A}{\partial p_j} \right)$$

quantum

$$(A, B) = \frac{1}{2i} \{ \hat{A}\hat{B} - \hat{B}\hat{A} \}$$

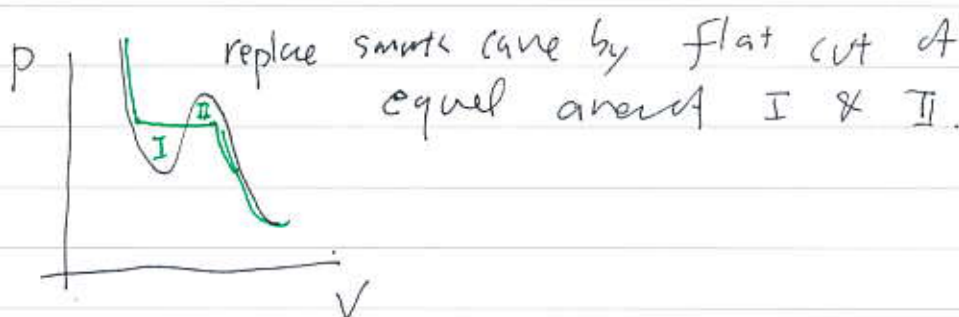
- (d) ergodic hypothesis:

time average = phase space average

$$\frac{1}{T} \int_0^T A(t) dt = \int d\Gamma A(\Gamma) \rho(\Gamma)$$

equilibrium distribution
no eq. -1

- (e) Maxwell construction



when $dV=0$

2. (a)
7

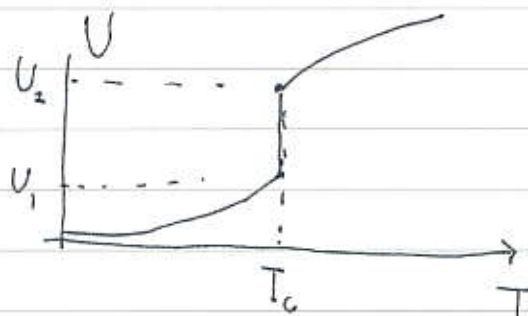
$$\frac{dS}{dU} = \frac{1}{T}$$

$$\frac{d^2S}{dU^2} = \frac{d}{dU} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{dT}{dU} \quad 331$$

$$\text{Date} = -\frac{1}{T^2 C}$$

$$U \quad L \quad -1$$

6 (b)



two phase coexist at T_c

6 (c)

1st order phase transition point at T_c , U_1 jumps to U_2 as T increases
 $C \quad \frac{\partial^2 S}{\partial U^2} = 0 \quad C \rightarrow +\infty$ U_1 is initial energy of pure phase 1

6

(d)

$$dQ = TdS \rightarrow \delta Q = T_c (S_2 - S_1) = U_2 - U_1$$

$$S = \frac{1}{T_c} U + \text{const}$$

$$S_1 = \frac{1}{T_c} U_1 + \text{const}$$

$$S_2 = \frac{1}{T_c} U_2 + \text{const}$$

$$S_2 - S_1 = \frac{1}{T_c} (U_2 - U_1)$$

$$\beta p^2 = x^2$$

15 mark

3. (a)

$$Z = \frac{1}{h^2} \int dp_x \int dp_y \int dx \int dy e^{-\beta \left(\frac{1}{2} p_x^2 + \frac{1}{2} p_y^2 + \frac{1}{2} (x^2 + y^2 + xy) \right)}$$

10 (b)
mark

$$= \frac{1}{h^2} \left[\int_{-\infty}^{\infty} dp e^{-\beta \frac{p^2}{2}} \right]^2 \int dx \int dy e^{-\frac{\beta}{2} (x^2 + y^2 + xy)}$$

$$= \frac{1}{h^2} \left[\frac{1}{\sqrt{\beta}} \int_{-\infty}^{\infty} dx e^{-\frac{\beta x^2}{2}} \right]^2 \int dx \int dy e^{-\frac{\beta}{2} (x^2 + y^2 + xy)}$$

$$= \frac{1}{h^2 \beta} 2\pi \int_{-\infty}^{\infty} dx e^{-\frac{\beta x^2}{2}} \int_{-\infty}^{\infty} dy e^{-\frac{\beta}{2} (y^2 + xy)}$$

$$= \frac{2\pi}{h^2 \beta} \int_{-\infty}^{\infty} dx e^{-\frac{\beta x^2}{2}} \int_{-\infty}^{\infty} dy e^{-\frac{\beta}{2} \left[\left(y + \frac{x}{2} \right)^2 - \frac{x^2}{4} \right]}$$

$$3) = \frac{2\pi}{h^2 \beta} \int dx e^{-\frac{\beta x^2}{2}} e^{+\frac{\beta x^2}{8}} \underbrace{\int dy' e^{-\frac{\beta y'^2}{2}}}_{\frac{\sqrt{2\pi}}{\beta}} \quad y' = y + \frac{x}{2} \quad 2333$$

$$= \frac{2\pi}{h^2 \beta} \sqrt{\frac{2\pi}{\beta}} \int dx e^{-\frac{3\beta x^2}{8}}$$

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$$-\frac{1}{2} + \frac{1}{8} = -\frac{4}{8} + \frac{1}{8}$$

$$= -\frac{3}{8}$$

$$Z = \frac{2\pi}{h^2 \beta} \sqrt{\frac{2\pi}{\beta}} \sqrt{\frac{2\pi}{3\beta/4}} = \frac{(2\pi)^2 \sqrt{4/3}}{h^2 \beta^2} = \frac{8\pi^2}{h^2 \beta^2 \sqrt{3}} = -\frac{3}{8}$$

$$F = -\frac{1}{\beta} \ln Z = -\frac{1}{\beta} \ln \left(\frac{(2\pi)^2 \sqrt{4/3}}{h^2 \beta^2} \right) = \frac{2}{\beta} (\ln \beta + \text{const})$$

correct former incorrect select -7

$$e^{-\beta(H - \mu N)}$$

4.

$$(a) \quad Z = \sum_{N=0}^{\infty} e^{-\beta(-(N-1)\epsilon - \mu N)}$$

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$$= \sum_{N=0}^{\infty} e^{\beta(\epsilon + \mu)N} e^{-\beta\epsilon}$$

$$= \frac{e^{-\beta\epsilon}}{1 - e^{\beta(\epsilon + \mu)}}$$

$$(b) \quad \psi = -\frac{1}{\beta} \ln Z = -\frac{1}{\beta} \ln \left[\frac{e^{-\beta\epsilon}}{1 - e^{\beta(\epsilon + \mu)}} \right]$$

6

$$= -\frac{1}{\beta} \left[\ln e^{-\beta\epsilon} - \ln(1 - e^{\beta(\epsilon + \mu)}) \right]$$

$\langle N \rangle > 0$

$\epsilon + \mu < 0$

$$= \epsilon + \frac{1}{\beta} \ln(1 - e^{\beta(\epsilon + \mu)})$$

7

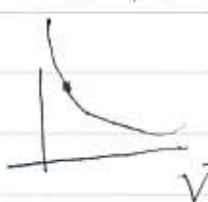
$$(c) \quad \langle N \rangle = -\frac{\partial \psi}{\partial \mu} = -\frac{1}{\beta} \frac{-e^{\beta(\epsilon + \mu)} \cdot \beta}{1 - e^{\beta(\epsilon + \mu)}} = \frac{1}{e^{-\beta(\epsilon + \mu)} - 1}$$

$$\begin{aligned}
 \int e^{-\beta a r^3} 4\pi r^2 dr &= \int e^{-a r^3} 4\pi \frac{dr^3}{3} \\
 &= \frac{4\pi}{3a} \int_0^{\infty} e^{-a r^3} d a r^3 \\
 e^{-\beta H} \quad H = p + a &= \frac{4\pi}{3a} e^{-a r^3} \Big|_0^{\infty} \\
 &= \frac{4\pi}{3a} [1 - e^{-\beta a R^3}]
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{1} \quad Z &= \beta^{\frac{3N}{2}} (1 - e^{-a R^3})^N \quad a \text{ small} \\
 &= \beta^{\frac{3N}{2}} (1 - 1 + a R^3)^N \\
 &= \beta^{\frac{3N}{2}} (1 - e^{-a' V})^N
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad F &= -\frac{N}{\beta} \ln(1 - e^{-a' V}) + f(T) \\
 \Rightarrow P &= -\frac{\partial F}{\partial V} = \frac{N}{\beta} \frac{-e^{-a' V} (-a')}{1 - e^{-a' V}} = \frac{N}{\beta} \frac{a'}{e^{a' V} - 1}
 \end{aligned}$$

$a' > 0$



$$\begin{aligned}
 U &= -\frac{\partial \ln Z}{\partial \beta} = -\frac{\partial}{\partial \beta} \left[\frac{3N}{2} \ln \beta + V \right] = T \\
 \Delta S &= \int \frac{dQ}{T} \quad \text{conditional} \\
 &= \int \frac{dQ}{T} \quad U(\beta, V) \\
 a R^3 &= \left[\frac{4\pi}{3a} R^3 \right] = \frac{4\pi}{3} \frac{R^3}{V_0} = \frac{U}{V_0}
 \end{aligned}$$

$$\Rightarrow N = -\frac{\partial F}{\partial T} = \frac{\partial f(T)}{\partial T} + N(1 - e^{-a V}) + \frac{1}{1 - e^{-a V}}$$

$$\Delta S = -\frac{\partial F}{\partial \beta} \frac{\partial \beta}{\partial T} = \frac{1}{T} (-\beta^2)$$

expression

$$\delta Q = 0, \quad \delta W = 0$$

$$e^x = 1 + x + \frac{x^2}{2} + \dots$$

$$\int dU = \delta Q - \delta W = 0$$

Date

$$Z = \beta^{\frac{3N}{2}} \left(1 - \left(1 - a'V + \frac{a'^2}{2} V^2 \right) \right)^N$$

$$= \beta^{\frac{3N}{2}} \left(a'V + \frac{a'^2}{2} V^2 \right)^N$$

$$= \beta^{\frac{3N}{2}} \left(V + \frac{a'}{2} V^2 \right)^N$$

$$a' = \beta a$$

$$F = -\frac{1}{\beta} \ln Z = -\frac{1}{\beta} \left[\frac{3N}{2} \ln \beta + N \ln \left(V + \frac{a'}{2} V^2 \right) \right]$$

So

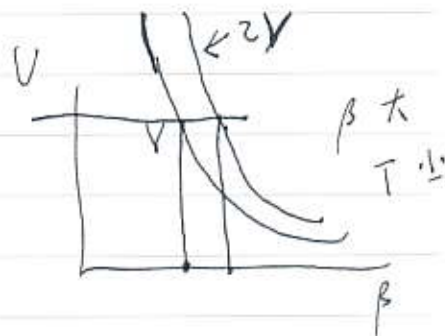
$$= + kT \frac{3N}{2} \ln T - NkT \ln V - NkT \frac{a'}{2} V$$

$$= + kT \frac{3N}{2} \ln T - NkT \ln V - \frac{N}{2} V a$$

$$\frac{\partial S}{\partial T} = \frac{3N}{2} \ln T + \frac{3N}{2} k - NkT \ln V$$

or

$$Z = \beta^{\frac{3N}{2}} (1 - e^{-\beta a V})^N \cdot \frac{1}{\beta a}$$



$$\left[\int e^{-\beta p^2} dp \right]^3 \int e^{-\beta a r^3} r^2 dr$$

$$\left(\frac{1}{\sqrt{\beta}} \right)^3 \frac{1}{\beta} \int e^{-\beta a r^3} dr^3 \beta$$

$$\beta^{-\frac{3}{2}-1} \left[e^{-\beta a r^3} \right]_0^R$$

$$\beta^{-\frac{5}{2}} [1 - e^{-\beta a V}]$$

QE 08

$$H = \sum_{i=1}^N \left(\frac{\vec{p}_i^2}{2m} + V(\vec{r}_i) \right)$$

$$V(r) = \begin{cases} ar^3 & r < R \\ +\infty & r > R \end{cases} \quad a > 0$$

Date

(a)
+3

$$Z = Z^N$$

$$Z = \frac{1}{h^3} \int_{-\infty}^{\infty} d\vec{p} e^{-\beta \frac{\vec{p}^2}{2m}} \int d\vec{r} e^{-\beta V(\vec{r})}$$

$$= \frac{1}{h^3} \left(\frac{m}{\beta} \right)^3 \int_0^R 4\pi r^2 dr e^{-\beta ar^3} \quad (r^3)' = 3r^2$$

$$= \frac{1}{h^3} \left(\frac{m}{\beta} \right)^{\frac{3}{2}} 4\pi \int_0^R e^{-\beta ar^3} d\frac{r^3}{3} \quad (\beta a) \frac{1}{\beta a}$$

$$= \frac{1}{h^3} \left(\frac{m}{\beta} \right)^{\frac{3}{2}} \frac{4\pi}{3} \frac{1}{\beta a} \int_0^R e^{-\beta ar^3} d(\beta ar^3) \quad V = \frac{4\pi R^3}{3}$$

$$= \frac{1}{h^3} \left(\frac{m}{\beta} \right)^{\frac{3}{2}} \frac{4\pi}{3} \frac{1}{\beta a} \left[1 - e^{-\beta a R^3} \right]$$

$$\int_0^{\infty} e^{-x} dx = -e^{-x} \Big|_0^{\infty}$$

$$= \text{const } T^{\frac{5}{2}} \left[1 - e^{-\frac{a}{k_B T} \frac{3}{4\pi} V} \right]$$

(b)

+3

$$P = \frac{k_B T}{\beta} \frac{\partial \ln Z}{\partial V}$$

$$F = -S_0 T - p dV$$

$$F = -k_B T \ln Z$$

$$= k_B T \frac{\partial}{\partial V} \left[\frac{5}{2} \ln T + \ln \left(1 - e^{-\frac{a}{k_B T} \frac{3}{4\pi} V} \right) \right] \cdot N \quad \frac{\partial F}{\partial V} = -p$$

$$= k_B T N \frac{e^{-\frac{a}{k_B T} \frac{3}{4\pi} V}}{1 - e^{-\frac{a}{k_B T} \frac{3}{4\pi} V}} \cdot \frac{a}{k_B T} \frac{3}{4\pi}$$

$$= \frac{3a}{4\pi} N \frac{1}{e^{\frac{3aV}{4\pi k_B T}} - 1}$$

if $a \rightarrow 0$

$$\left(\frac{3a}{4\pi} \right) \frac{1}{\left(\frac{3a}{4\pi} \right) \frac{V}{k_B T}} = \frac{k_B T N}{V}$$

(c)

+2

L

correct for adiabatic quasi static process

$$dS = \frac{\delta Q}{T} \quad \delta Q = 0 \rightarrow \Delta S = 0$$

Date

(d) T decrease. $\Delta Q = 0, \Delta W = 0 \rightarrow \Delta U = 0$

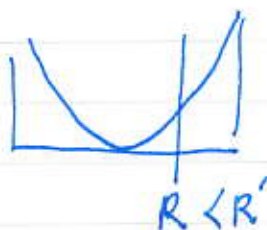
+2

by 1st law of thermodynamics

$$\Delta U = \Delta Q + \Delta W = 0$$

$$U = \left\langle \frac{1}{2} m \vec{v}^2 \right\rangle + \langle v \rangle = \text{const}$$

clearly $\langle v \rangle$ is bigger so $\left\langle \frac{1}{2} m \vec{v}^2 \right\rangle$ must be smaller $T \downarrow$.



$$PV = NkT$$

ideal gas adiabatic process $S \sim \ln(V T^{\frac{3}{2}})$

$$\sim \ln(V^{\frac{5}{2}} P^{\frac{1}{2}})$$

Δ

$$dU = Tds - pdv$$

$$PV^{\frac{5}{3}} = \text{const}$$

$$dS = \frac{dU + pdv}{T} = \frac{C_v dT + \frac{\lambda}{V^{\frac{5}{3}}} dV}{T} = 0$$

Langevin Eq. by Fourier method

Date

$$\frac{dv}{dt} = -\gamma v + \frac{R(t)}{m}$$

$$\langle R(t) \rangle = 0$$

$$\langle R(t) R(t') \rangle = C \delta(t-t')$$

$$\tilde{R}[\omega] = \int_{-\infty}^{\infty} R(t) e^{+i\omega t} dt$$

$$\tilde{v}[\omega] = \int_{-\infty}^{\infty} v(t) e^{+i\omega t} dt$$

$$v(t) = \int_{-\infty}^{\infty} \tilde{v}[\omega] e^{-i\omega t} \frac{d\omega}{2\pi}$$

$$\frac{dv}{dt} = \int_{-\infty}^{\infty} \tilde{v}[\omega] (i\omega) e^{-i\omega t} \frac{d\omega}{2\pi}$$

$$\rightarrow -i\omega \tilde{v}[\omega] + \gamma \tilde{v}[\omega] = \frac{\tilde{R}[\omega]}{m}$$

$$\rightarrow \tilde{v}[\omega] = \frac{\tilde{R}[\omega]}{m(i\omega + \gamma)}$$

$$I_v[\omega] = \frac{I_R[\omega]}{m^2(\omega^2 + \gamma^2)}$$

$$\rightarrow |\tilde{v}[\omega]|^2 = \frac{|\tilde{R}[\omega]|^2}{m^2(\omega^2 + \gamma^2)}$$

← Wiener-Kinchin theorem

$$\langle \tilde{v}[\omega] \tilde{v}^{*}[\omega'] \rangle = I_v[\omega] \delta(\omega - \omega')$$

$$\text{where } I_v[\omega] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \langle v(t) v(t') \rangle e^{-i\omega t} dt$$

$$\text{Similarly } \langle \tilde{R}[\omega] \tilde{R}^{*}[\omega'] \rangle = I_R[\omega] \delta(\omega - \omega') \cdot 2\pi$$

$$I_R[\omega] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{\langle R(t) R(t') \rangle}_{C \delta(t)} e^{+i\omega t} dt = \frac{C}{2\pi}$$

$$\langle |\tilde{v}[\omega]|^2 \rangle = I_v[\omega] \delta(0) \cdot 2\pi \rightarrow I_v[\omega] \delta(0) = \frac{I_R[\omega] \delta(0)}{m^2(\omega^2 + \gamma^2)}$$

$$\langle |\tilde{R}[\omega]|^2 \rangle = I_R[\omega] \delta(0) \cdot 2\pi \rightarrow I_v[\omega] = \frac{I_R[\omega]}{m^2(\omega^2 + \gamma^2)}$$

$$I_v(\omega) = \frac{C/(2\pi)}{m^2(\omega^2 + \gamma^2)}$$

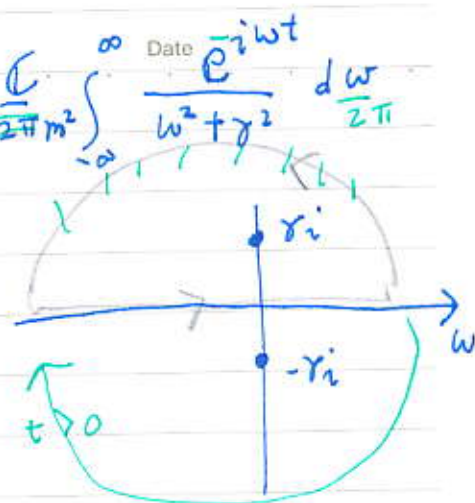
$$\omega^2 + \gamma^2 = 0 \quad \omega = \pm \gamma i \quad 345$$

$$\langle v(t) v(u) \rangle = \int_{-\infty}^{\infty} I_v(\omega) e^{i\omega t} d\omega = \frac{C}{2\pi m^2} \int_{-\infty}^{\infty} \frac{e^{i\omega t}}{\omega^2 + \gamma^2} \frac{d\omega}{2\pi}$$

Residue theorem

$$= \frac{C}{2\pi m^2} \left[\int_{-R}^R d\omega + \int_{\omega=R e^{i\theta}} d\omega \right]$$

$\omega = R e^{i\theta}$
 $\theta \in [0, \pi]$



$$= \frac{C}{2\pi m^2} \left(\text{Residue at } \gamma i \right)$$

$$\left. \frac{(\omega + i\gamma) e^{i\omega t}}{(\omega + i\gamma)(\omega - i\gamma)} \right|_{\omega = i\gamma}$$

$$= \frac{C}{2\pi m^2} \cdot \frac{e^{-\gamma t}}{2i\gamma}$$

$$\gamma_i = \gamma i$$

$$\omega \rightarrow i\gamma$$

$$= \frac{C e^{-\gamma t}}{2m^2 \gamma}$$

$$t > 0$$

$$= \frac{e^{-\gamma t}}{2i\gamma}$$

$$\omega = i\gamma$$

$$= \frac{K_B T}{m} e^{-\gamma t}$$

$$C = 2m K_B T$$

$$\frac{a-1}{z-z_0}$$

residue theorem

$$\oint f(z) dz = 2\pi i \sum_j \text{Res}(f, z_j)$$

in side
right hand
path

$$\text{Res at } z_j \text{ is } \frac{a-1}{z-z_j}$$

Q2. Find ex

Date 6 May 08

$$H = \sum_{i=1}^N \mathcal{E}_i$$

$$\mathcal{E}_i = \frac{p_i^2}{2m} + u(r_i)$$

$$\mathbb{Z} = \mathcal{Z}^N$$

(a) Yes, $T_1 = T_2 = T$ (b) canonical. since total particle number N is fixed.

$$(c) e^{\beta \left(\frac{p^2}{2m} + u(r_i) \right)} d\vec{p} d\vec{r}$$

$$N_1 + N_2 = N$$

$$N_1 = N_2 e^{\beta u_0}$$

$$\frac{N_1}{N_2} = \frac{1}{e^{-\beta u_0}} = e^{\beta u_0}$$

$$N_2 (1 + e^{\beta u_0}) = N$$

$$N_2 = \frac{N}{1 + e^{\beta u_0}}$$

$$N_1 = \frac{N e^{\beta u_0}}{1 + e^{\beta u_0}} = \frac{N}{1 + e^{-\beta u_0}}$$

$$N_1 = N \frac{1}{1 + e^{\beta u_0}}$$

$$N_1 + N_2 = N \left[\frac{1}{1 + \frac{1}{x}} + \frac{1}{1+x} \right]$$

$$N_2 = N \frac{1}{1 + e^{\beta u_0}}$$

$$= N \left[\frac{x}{x+1} + \frac{1}{1+x} \right]$$

$$\frac{N_1}{N_2} = \frac{\frac{1}{x + \frac{1}{x}}}{\frac{1}{1+x}} = (x)$$

$$= N \left[\frac{1+x}{1+x} \right] = N$$

$$x = e^{\beta u_0}$$

$$z = C(T) [v_1 + v_2 e^{-\beta u_0}]$$

$$F = -k_B T \ln [v_1 + v_2 e^{-\beta u_0}] + C'(T)$$

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$$p_1 = -\frac{\partial F}{\partial v_1} = - \left(-k_B T N \frac{1}{v_1 + v_2 e^{-\beta u_0}} \right) = \frac{k_B T N}{v(1 + e^{-\beta u_0})}$$

$$p_2 = k_B T N \frac{e^{-\beta u_0}}{v_1 + v_2 e^{-\beta u_0}} = \frac{k_B T N}{v(1 + e^{\beta u_0})}$$

$$\begin{bmatrix} e^{\kappa - \lambda} & 1 & 1 \\ 1 & e^{\kappa - \lambda} & 1 \\ 1 & 1 & e^{\kappa - \lambda} \end{bmatrix} = \begin{bmatrix} \alpha + 1 & 1 & 1 \\ 1 & \alpha + 1 & 1 \\ 1 & 1 & \alpha + 1 \end{bmatrix} = (\alpha + 1)^3 + 2 - 3(\alpha + 1) = 0$$

$$(\alpha + 1)^3 - 3\alpha - 1 = 0$$

$$\alpha^3 + 3\alpha^2 + 3\alpha + 1 - 3\alpha - 1 = 0$$

$$\alpha(\alpha^2 + 3\alpha + 6) = 0$$

$$\alpha^3 + 3\alpha^2 = 0$$

$$\alpha^2(\alpha + 3)$$

220

$$\lambda = e^{\kappa - 1} - \alpha$$

Q4

$$(a) \quad \chi = \beta \int_0^L dr \, G(r)$$

$$= \beta \int_0^L r^{d-1} dr \frac{1}{r^{d-2+\eta}} = \int_0^L dr \cdot r^{1-\eta} \quad r^{1-\eta+1} = L^{2-\eta}$$

$$\chi = \frac{\partial m}{\partial h}$$

$$r^{-1+\eta}$$

$$m = \frac{\partial f}{\partial h}$$

$$f(t, h, L) = b^{-d} f(b^x t, b^x h, \frac{b}{L})$$

$$\chi = -\frac{\partial^2 f}{\partial h^2} = -b^{-d} \frac{\partial^2 f}{\partial h^2} (b^x t, b^x h, \frac{b}{L})$$

$$= -f''(t, h, L) = -b^{-d} b^{2x} \frac{\partial^2 f}{\partial (b^x h)^2} (b^x t, b^x h, \frac{b}{L})$$

$$\chi(t, h, L) = b^{2x-d} \chi(b^x t, b^x h, \frac{b}{L})$$

$$a = 2x-d \quad \gamma = \frac{2x-d}{2}$$

$$= y \gamma$$

$$y = \frac{1}{2}$$

$$= \frac{\gamma}{2}$$

$$f(z) = \frac{1}{z^2} \left(\frac{1}{z} + \frac{1}{z} \right) = \frac{2}{z^3}$$

$$\frac{d}{dz} \left(\frac{1}{z^2} \right) = -\frac{2}{z^3}$$

$$\frac{d^2}{dz^2} \left(\frac{1}{z^2} \right) = \frac{6}{z^4}$$

$$\left(\frac{1}{z^2} \right)' = -\frac{2}{z^3}$$

$$\left(\frac{1}{z^2} \right)'' = \frac{6}{z^4}$$

$$\left(\frac{1}{z^2} \right)''' = -\frac{24}{z^5}$$

$$\left(\frac{1}{z^2} \right)^{(4)} = \frac{240}{z^6}$$

$$\frac{d^4}{dz^4} \left(\frac{1}{z^2} \right) = \frac{240}{z^6}$$

$$\oint_C f(z) dz = 2\pi i (\text{sum of enclosed residues})$$

$$\text{residue is } a_{-1} \text{ of } f(z) = \dots + \frac{a_{-1}}{z-z_0} + \dots$$

After contour integral we get

$$F(t) = \frac{C}{4m^2} \left[\frac{e^{-\frac{\gamma}{2}t} \cdot (-i\sqrt{\frac{\kappa}{m} - \frac{\gamma^2}{4}}t)}{\gamma\sqrt{\frac{\kappa}{m} - \frac{\gamma^2}{4}}} \cdot \left(i\frac{\gamma}{2} - \sqrt{\frac{\kappa}{m} - \frac{\gamma^2}{4}} \right) + \text{c.c.} \right]$$

Q5.

$$m \frac{d^2 x}{dt^2} = -kx - m\gamma \frac{dx}{dt} + R(t)$$

[a]

$$x(t) = \int_{-\infty}^{\infty} \tilde{x}(\omega) e^{-i\omega t} \frac{d\omega}{2\pi}$$

$$(-i\omega)^2 m \tilde{x} = -k \tilde{x} - m\gamma (-i\omega) \tilde{x} + \tilde{R}$$

$$\rightarrow \tilde{x} = \frac{\tilde{R}}{k + im\gamma\omega - \omega^2 m} = \frac{\tilde{R}(\omega)}{k - im\gamma\omega - \omega^2 m}$$

(b)

$$\tilde{F}(\omega) = \frac{C}{|k - im\gamma\omega - \omega^2 m|^2} = \frac{C}{(K - \omega^2 m)^2 + (m\gamma\omega)^2}$$

$$\begin{aligned} \langle \tilde{x}(\omega) \tilde{x}(\omega')^* \rangle &= \tilde{F}(\omega) \cdot 2\pi \delta(\omega - \omega') \\ &= \frac{1}{K - im\gamma\omega - \omega^2 m} \cdot \frac{1}{(K - im\gamma\omega' - \omega'^2 m)^*} \langle \tilde{R}(\omega) \tilde{R}(\omega')^* \rangle \end{aligned}$$

← Four transform of real time

$$\langle \int_{-\infty}^{\infty} R(t) R(t') e^{i\omega(t-t')} dt \cdot 2\pi \delta(\omega - \omega') \rangle$$

$C \delta(t-t')$

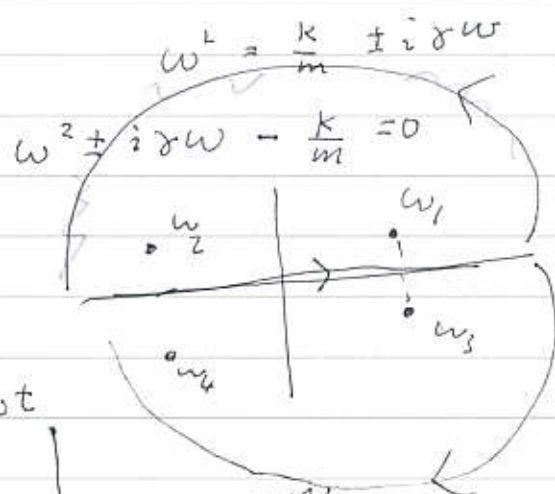
$$\tilde{F}(\omega) = \frac{C}{|k - im\gamma\omega - \omega^2 m|^2}$$

$$k - \omega^2 m = \pm i m \gamma \omega$$

$$(c) \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{C e^{-i\omega t}}{|k - im\gamma\omega - \omega^2 m|^2} d\omega$$

$$m\omega^2 = K \pm im\gamma\omega$$

$$\omega = \frac{\pm i\gamma \pm \sqrt{-\gamma^2 + 4 \frac{K}{m}}}{2}$$



assume
 $t > 0$

$$F(t) = 2\pi i \text{Res} \left[\frac{C}{|z|^2} e^{i\omega t} \right]_{\text{at } \omega_1, \omega_2}$$

$$(\omega - \omega_1)(\omega - \omega_2)(\omega - \omega_3)(\omega - \omega_4)$$

each ph is

K Huang p. 8.5

Date 26 Feb 09

grand with respect to occupation # (# of photons)

$$\Omega = \sum_{n=0,1,2,\dots} (e^{\beta \mu})^n Q_n$$

$$Q_n = \sum e^{-\beta E_n} \quad \uparrow \text{ energy of } n \text{ photons}$$

$$E_n = \hbar \omega n$$

$$= g e^{-\beta \hbar \omega n}$$

 g_n degeneracy

$$\mathcal{H} = \sum_j \hat{a}_j^\dagger \hat{a}_j \hbar \omega$$

 n photon state

$$\hat{a}_j^\dagger \hat{a}_j^\dagger \dots \hat{a}_j^\dagger |0\rangle$$

create n -times

wave for

$$\phi(x_1) \phi(x_2) \phi(x_3)$$

$n_1 \quad n_2 \quad n_3$

$$n = \sum_{j=1}^N n_j$$

 $\hat{a}$

$$g_n = \sum_{\substack{n_j \\ \sum n_j = n}} 1$$

$$\text{or } \sum e^{-\beta \hbar \omega n_j} = e^{-\frac{1}{2} \beta \hbar \omega N}$$

$$\Omega = \sum_{n=0,1,2,\dots} \sum_{\sum n_j = n} 1 e^{-\beta \hbar \omega n}$$

$$= \sum_{n_1, n_2, \dots, n_N} e^{-\beta \hbar \omega \sum n_j} = \left(\sum_{n_j=0}^{\infty} e^{-\beta \hbar \omega n_j} \right)^N =$$

$$= \left(\frac{1}{1 - e^{-\beta \hbar \omega}} \right)^N \cdot e^{-\beta \frac{1}{2} \hbar \omega N}$$

$\hbar \rightarrow 0$
 $\omega \rightarrow 0$

 N is fixedthis gives N oscillators labelled by j $j=1, 2, \dots, N$ for the n th oscillatorthe # of photons or photons is n_j in oscillator j

$$\sum_j n_j = n \text{ give the total}$$

$$\frac{1}{\beta \hbar \omega}$$

$$\frac{1}{1 - e^{-\beta \hbar \omega}}$$

$$= \frac{1}{1 - e^{-\beta \hbar \omega}} = \frac{1}{1 - e^{-\beta \hbar \omega}}$$

$$E = \sum_{k=0}^{\infty} \epsilon_k n_k \quad N = \sum_{k=0}^{\infty} n_k$$

↑ energy of 1 particle at level k

total # of particles

Each set $\{n_k\}$ is one state

$$\begin{aligned} \bar{Z} &= \sum_{\{n_k\}} e^{-\beta(E - \mu N)} = \sum_{\{n_k\}} e^{-\beta \sum_k (\epsilon_k - \mu) n_k} \\ &= \prod_k \sum_{n_k=0}^{\infty} \left(e^{-\beta(\epsilon_k - \mu)} \right)^{n_k} = \prod_k \frac{1}{1 - e^{-\beta(\epsilon_k - \mu)}} \end{aligned}$$

Boltzmann sum

$$\begin{aligned} \bar{Z}_B &= \sum_{N=0}^{\infty} \sum_{\{n_k\}} \frac{1}{n_1! n_2! n_3! \dots} e^{-\beta(E_N - \mu N)} \\ &= \sum_{\{n_k\}} \prod_{k=0}^{\infty} \frac{[e^{-\beta(\epsilon_k - \mu)}]^{n_k}}{n_k!} = \prod_{k=0}^{\infty} e^{[e^{-\beta(\epsilon_k - \mu)}]} \\ &= e^{\sum_{k=0}^{\infty} e^{-\beta \epsilon_k} e^{\beta \mu}} \\ &= \exp \left[e^{\beta \mu} e^{-\beta \hbar \omega_0 / 2} \frac{1}{1 - e^{-\beta \hbar \omega_0}} \right] \end{aligned}$$

$$\bar{Z} = \sum_{N=0}^{\infty} \frac{\bar{Z}^N}{N!} \bar{Z}_1^N$$

gives same answer

$\bar{Z}_1$ 1 particle partition function

for each given set $\{n_j\}$ the degeneracy is 1

For Boltzmann $g\{n_j\} = \frac{1}{\prod_j n_j!} \leftarrow$ exchange the phonon and different is used then

$$\begin{aligned} \Gamma_B &= \sum_{n=0,1,2,\dots,N} \sum_{\substack{\sum n_j = n \\ j=1}}^N \frac{e^{-\beta \hbar \omega (\sum n_j + \frac{1}{2})}}{\prod_j n_j!} N! \\ &= \sum_{n_1, n_2, \dots} \frac{e^{-\beta \hbar \omega n_1}}{n_1!} \frac{e^{-\beta \hbar \omega n_2}}{n_2!} \dots = \left(\sum_{n_j} \frac{e^{-\beta \hbar \omega (n_j + \frac{1}{2})}}{n_j!} \right)^N \\ &= e^{-\beta \hbar \omega \frac{N}{2}} \left[\sum_{n_j=0}^{\infty} \frac{(e^{-\beta \hbar \omega})^{n_j}}{n_j!} \right]^N \\ &= e^{-\beta \hbar \omega \frac{N}{2}} \exp[N e^{-\beta \hbar \omega}] \end{aligned}$$

when $\beta \rightarrow 0$

$$e^{-\beta \hbar \omega} = e^{(1 - \beta \hbar \omega + \dots)} = e^{-\beta \hbar \omega}$$

$$Q_n = \frac{1}{N!} (e^{-\beta \hbar \omega})^N$$

$$\Gamma_B = \sum_{n=0}^{\infty} \frac{(N e^{-\beta \hbar \omega})^n}{n!} = e^{N e^{-\beta \hbar \omega}}$$

If we have

$$\begin{aligned} \Gamma_B &= \sum_{n=0,1,2,\dots} \sum_{\substack{\sum n_j = n \\ j=1}}^N \frac{n! e^{-\beta \hbar \omega \sum n_j}}{\prod_j n_j!} e^{-\beta \hbar \omega \frac{1}{2} N} \\ &= \sum_{n=0,1,2,\dots} \left(\sum_{j=1}^N e^{-\beta \hbar \omega} \right)^n = \sum_{n=0,1,2,\dots} (N e^{-\beta \hbar \omega})^n \\ &= \frac{1}{1 - N e^{-\beta \hbar \omega}} < \infty \quad \text{not correct} \end{aligned}$$

binomial

$$(x+y)^n = \sum_{k=0}^n \frac{n!}{k!(n-k)!} x^k y^{n-k}$$

Fokker-Planck equations new derivation

Date 26 Mar 09

$$\frac{dv}{dt} = -\gamma v + \frac{R(t)}{m} \quad \text{standard Langevin eq.}$$

for a given realization of the random noise $R(t)$

P : probability of finding the Brownian particle at velocity v to $v + dv$ is $P(v, t) dv$ at time t

$$\int_{-\infty}^{\infty} P(v, t) dv = 1 \quad \text{conservation of probability}$$

prob. is conserved $P \propto e^{-\frac{1}{2} m \gamma v^2}$

$$\frac{\partial P}{\partial t} + \nabla \cdot \vec{j} = 0$$

$$\vec{j} = \vec{v} P$$

$$\frac{\partial P(v, t)}{\partial t} + \frac{\partial}{\partial v} (\underbrace{\dot{v} P}_{\text{prob. flux}}) = 0$$

think of v as "x"

$$\frac{\partial P(v, t)}{\partial t} + \frac{\partial}{\partial v} \left(\left[-\gamma v + \frac{R(t)}{m} \right] P \right) = 0$$

 $\langle P \rangle$

original

not

is the first

is the first

formally the same as equation for v

$$\frac{\partial P(v, t)}{\partial t} = -\gamma \frac{\partial}{\partial v} (v P) + \frac{\partial}{\partial v} \left(\frac{R}{m} P \right)$$

$$= -\hat{L} P + \frac{\partial}{\partial v} \left(\frac{R}{m} P \right)$$

$$\frac{\partial P}{\partial t} \neq -\hat{L} P = \gamma \frac{\partial}{\partial v} (v P)$$

operator $\hat{L}$ acts on P

$$P(v, t) = e^{-t \hat{L}} P(v, 0) \quad \text{if } R = 0$$

$$P(v, t) = e^{-t \hat{L}} P(v, 0) = \int_0^t dt e^{-(t-\tau) \hat{L}} \frac{\partial}{\partial v} \left(\frac{R(\tau)}{m} P \right)$$

the is τ

$$P(v, t) = e^{-t\hat{L}} P(v, 0) - \int_0^t d\tau e^{-(t-\tau)\hat{L}} \frac{\partial}{\partial v} \left(\frac{R(\tau)}{m} P(v, \tau) \right) \quad 361$$

Date

iterate

$$\begin{aligned} \frac{\partial P(v, t)}{\partial t} &= -\hat{L} P(v, t) - \frac{\partial}{\partial v} \left(\frac{R(t)}{m} P(v, t) \right) \\ &= -\hat{L} P(v, t) - \frac{\partial}{\partial v} \left(\frac{R(t)}{m} e^{-t\hat{L}} P(v, 0) \right) \leftarrow \text{disturb} \\ &\quad + \frac{\partial}{\partial v} \left(\frac{R(t)}{m} \int_0^t d\tau e^{-(t-\tau)\hat{L}} \frac{\partial}{\partial v} \left(\frac{R(\tau)}{m} P(v, \tau) \right) \right) \end{aligned}$$

Avg. over noise $\langle R \rangle = 0$

$$\langle R(t) R(\tau) \rangle = C \delta(t - \tau)$$

wick the pair of $R(t)$ with P is zero.

$$\begin{aligned} \frac{\partial \langle P(v, t) \rangle}{\partial t} &= -\hat{L} \langle P(v, t) \rangle + \frac{\partial}{\partial v} \frac{1}{m^2} \int_0^t d\tau e^{-(t-\tau)\hat{L}} \underbrace{\langle R(t) R(\tau) \rangle}_{C \delta(t-\tau)} \frac{\partial \langle P(v, \tau) \rangle}{\partial v} \\ &= -\hat{L} \langle P(v, t) \rangle + \frac{C}{2m^2} \frac{\partial^2 \langle P(v, t) \rangle}{\partial v^2} \end{aligned}$$

why $\langle R(t) R(\tau) P(v, \tau) \rangle$

$$= \langle R(t) R(\tau) \rangle \langle P(v, \tau) \rangle ?$$

omit $\langle \rangle$

$$\rightarrow \frac{\partial P(v, t)}{\partial t} = \gamma \frac{\partial}{\partial v} (v P(v, t)) + \frac{C}{2m^2} \frac{\partial^2 P(v, t)}{\partial v^2}$$

$$C = 2m\gamma k_B T$$

$$\frac{C}{2m^2} = \frac{\gamma k_B T}{m}$$



P is a polynomial of noise $R(t)$

$$P(r, t) \dots \int_0^t d\tau R(\tau) \dots + \int_{\tau' < \tau} R(\tau') R(\tau) \dots$$

apply

Wick's theorem

$$\int R(\tau'') R(\tau') R(\tau) \dots$$

$\tau'' < \tau' < \tau$

$$\langle \underbrace{R(t) R(\tau)}_{\substack{\text{time } t < \tau \\ \text{in } P(r, \tau)}} P(r, \tau) \rangle \leftarrow \text{time } t < \tau \text{ in } P(r, \tau) \text{ is always smaller than } t$$

2 conditions needed for the validity of the derivation
 1) R is gaussian
 2) R is δ -correlated (white noise).

$$t < \tau \quad \tau' < \tau$$

$$R(t) R(\tau) R(\tau'') R(\tau')$$

$$\delta(t - t') = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{i\omega(t-t')} d\omega$$

if $|t - t'| \ll \tau$

$$t \geq \tau, \tau' \leq \tau$$

$$\tau \leq t, \tau'' \leq \tau'$$

$$\underbrace{R(t) R(\tau) R(\tau') R(\tau'')}_{\text{strictly non-increasing}}$$

$$\int \delta(t - \tau') \delta(\tau - \tau'') d\tau' d\tau''$$

if we keep $t > \tau$

Gaussian process $\langle R R \dots R \rangle = 0$ (odd #)

$$\langle \underbrace{R R \dots R}_{\text{even } n} \rangle = \sum \langle \dots \rangle \langle \dots \rangle$$

sum of product all possible pairings

we can assume $t_1 > t_2 > t_3 > t_4 > 0$ 365
if not we permute and rename
the variable

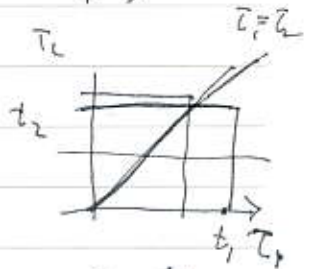
$$\langle \delta v(t_1) \delta v(t_2) \delta v(t_3) \delta v(t_4) \rangle$$

Date

$$= \left\langle e^{-\gamma t_1} \int_0^{t_1} \frac{R(\tau_1)}{m} e^{\gamma \tau_1} d\tau_1 e^{-\gamma t_2} \int_0^{t_2} \frac{R(\tau_2)}{m} e^{\gamma \tau_2} d\tau_2 \right. \\ \left. e^{-\gamma t_3} \int_0^{t_3} \frac{R(\tau_3)}{m} e^{\gamma \tau_3} d\tau_3 e^{-\gamma t_4} \int_0^{t_4} \frac{R(\tau_4)}{m} e^{\gamma \tau_4} d\tau_4 \right\rangle$$

3 terms

$$\frac{e^{-\gamma(t_1+t_2+t_3+t_4)}}{m^4} \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 e^{\gamma(\tau_1+\tau_2)} C \delta(\tau_1-\tau_2) \cdot$$



$$\int_0^{t_3} d\tau_3 \int_0^{t_4} d\tau_4 e^{\gamma(\tau_3+\tau_4)} C \delta(\tau_3-\tau_4)$$

+ permutations
12 34
[2 3 4]
[1 2 3 4]

$$= \frac{e^{-\gamma(t_1+t_2+t_3+t_4)}}{m^4} \int_0^{t_2} d\tau_2 e^{-2\gamma \tau_2} C^2 \int_0^{t_4} d\tau_4 e^{+2\gamma \tau_4}$$

$$\left[\frac{1}{2\gamma} (1 - e^{-2\gamma t_2}) \right] \frac{(1 - e^{-2\gamma t_4})}{2\gamma}$$

$$\frac{1}{2\gamma} (e^{2\gamma t_2} - 1) \frac{1}{2\gamma} (e^{2\gamma t_4} - 1)$$

$$= \frac{1}{(2\gamma)^2 m^4} \left[e^{-\gamma(t_1+t_2+t_3+t_4)} \left[e^{2\gamma(t_2+t_4)} - e^{2\gamma t_4} - e^{2\gamma t_2} + 1 \right] \right]$$

$$= \frac{1}{(2\gamma)^2 m^4} \left[e^{-\gamma(t_1-t_2+t_3-t_4)} - e^{-\gamma(t_1-t_2+t_3+t_4)} \right. \\ \left. - e^{-\gamma(t_1+t_2+t_3-t_4)} + e^{-\gamma(t_1+t_2+t_3+t_4)} \right]$$

or $\langle \delta v_1 \delta v_2 \delta v_3 \delta v_4 \rangle = \frac{C^2}{4\gamma^2 m^4} \left[e^{-\gamma|t_1-t_2|-\gamma|t_3-t_4|} \right. \\ \left. + e^{-\gamma|t_1-t_3|-\gamma|t_2-t_4|} \right. \\ \left. + e^{-\gamma|t_1-t_4|-\gamma|t_2-t_3|} \right]$

if $t_1 > t_2 > t_3 > t_4$ then

$$\langle \delta v_1 \delta v_2 \delta v_3 \delta v_4 \rangle = \frac{C^2}{4\gamma^2 m^4} \left[e^{-\gamma(t_1-t_2)-\gamma(t_3-t_4)} + e^{-\gamma(t_1-t_3)-\gamma(t_2-t_4)} \right. \\ \left. + e^{-\gamma(t_1-t_4)-\gamma(t_2-t_3)} \right]$$

Date 6 April 2010

Relation between 2 different form of Boltzmann Eq.
(c.f. S Harris 'an introduction to the Theory of the Boltzmann equation')

symmetric form

$$\text{mass} = 1$$

$$p = V$$

$$W(\vec{v}_1, \vec{v}_2, \vec{v}_1', \vec{v}_2') = \sigma(x, V) \delta(\vec{v}_1 + \vec{v}_2 - \vec{v}_1' - \vec{v}_2') \delta\left(\frac{V^2 - V'^2}{2}\right)$$

where $V = |\vec{v}_2 - \vec{v}_1|$ $V' = |\vec{v}_2' - \vec{v}_1'|$

W is my $\sigma(p, p_i | p', p'_i)$ on page 251. $\frac{V^2 - V'^2}{2}$

momentum conserved
energy conserved

x is scattering angle θ

$$\underbrace{d\vec{v}_1' d\vec{v}_2'}_{6\text{-dimensional integrals}} W(\vec{v}_1, \vec{v}_2, \vec{v}_1', \vec{v}_2')$$

$\vec{v}_1$ is parameter
 $\vec{v}_2$ last integral

consider variable change

$$\vec{v}_1', \vec{v}_2' \rightarrow \vec{V}', \vec{V}_c'$$

$$\vec{V}' = \vec{v}_2' - \vec{v}_1'$$

$$\vec{V}_c' = \frac{1}{2}(\vec{v}_1' + \vec{v}_2')$$

Jacob is $\begin{vmatrix} -1 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -1$

$$d\vec{V}_c' \delta(\dots - 2\vec{V}_c') = \frac{1}{2}$$

$$\text{so } d\vec{v}_1' d\vec{v}_2' = d\vec{V}' d\vec{V}_c'$$

$$\begin{aligned} \text{so } d\vec{v}_1' d\vec{v}_2' W(\vec{v}_1, \vec{v}_2, \vec{v}_1', \vec{v}_2') &= d\vec{V}' d\vec{V}_c' \sigma(x, V) \delta(\vec{v}_1 + \vec{v}_2 - \vec{v}_1' - \vec{v}_2') \delta\left(\frac{V^2 - V'^2}{2}\right) \\ &\quad \times \delta(\vec{v}_1 + \vec{v}_2 - 2\vec{V}_c') \delta\left(\frac{V^2 - V'^2}{2}\right) \text{ integr. on } d\vec{V}_c' \\ &= \frac{1}{2} d\vec{V}' \sigma(x, V) \delta\left(\frac{V^2 - V'^2}{2}\right) \\ &= \frac{1}{2} d\Omega V'^2 dV' \sigma(x, V) \delta\left(\frac{V^2 - V'^2}{2}\right) \\ &= \frac{1}{2} d\Omega V'^2 dV' \sigma(x, V) \delta\left(\frac{(V - V')(V + V')}{2}\right) \\ &\quad \frac{11}{V} \delta(V' - V) \end{aligned}$$

$4\pi R^2 dR$

$\delta(ax) = \frac{\delta(x)}{|a|}$

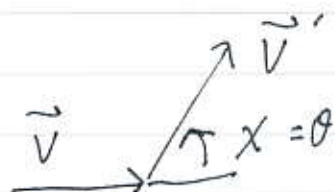
$V' = V$

why extra factor of 2.

$$d\vec{v}_1' d\vec{v}_2' W(\vec{v}_1, \vec{v}_2, \vec{v}_1', \vec{v}_2') = \frac{1}{2} d\Omega \cdot V \sigma(\chi, V) \quad \text{Date}$$

To avoid the factor 2 we should define

$$W(\vec{v}_1, \vec{v}_2, \vec{v}_1', \vec{v}_2') = 2 \sigma(\chi, V) \delta(\vec{v}_1 + \vec{v}_2 - \vec{v}_1' - \vec{v}_2') \delta\left(\frac{V^2 - V'^2}{2}\right)$$



Example of use of Green-Kubo

Ohmic law

(1) electrical conduction

$$\vec{J} = \sigma \vec{E}$$

(current density) ↑ applied E-field
electronic conductivity

$$H_1 = - \vec{P} \cdot \vec{E}(t)$$

dipole moment $\vec{P} = \sum_j e_j \vec{r}_j$

$$\hat{A} \text{ is } \vec{P} = \vec{A}$$

$$\vec{J} = \frac{\partial \vec{P}}{\partial t} = \sum_j e_j \vec{v}_j$$

$$\hat{B} \text{ is } \dot{\vec{P}} = \dot{\vec{A}} = \hat{B}$$

according to Green-Kubo

$$\langle \hat{B}(t) \rangle = + \beta \int_0^t dt' \langle \hat{B}_2(t); \hat{A}(t') \rangle a(t')$$

$$\langle \hat{J}(t) \rangle = \beta \int_0^t dt' \langle \hat{J}(t); \hat{J}(t') \rangle \vec{E}(t')$$

conductance tensor
current

thermal current → phonon conductivity

P_{xy} → viscosity η

pressure tensor

Date 25/3/11

linear response theory

$$H(t) = \hat{H}_0 - a(t) \hat{A}$$

↑
scalar field

wave function $|\psi\rangle$

e.g. a $\hat{A}$
 E-field dipole moment
 h field magnetization
 B is $\langle p(t) \rangle$
 polarization
 E field dipole moment

$$U(t, t') |\psi(t')\rangle = |\psi(t)\rangle$$

$$i\hbar \frac{\partial U(t, t')}{\partial t} = \hat{H}(t) U(t, t')$$

$$\vec{E} \cdot \vec{p}$$

$$\hat{\rho}(t) = \sum_i w_i |\psi_i(t)\rangle \langle \psi_i(t)| = U(t, 0) \hat{\rho}(0) U(0, t)$$

$$\rightarrow i\hbar \frac{\partial \hat{\rho}}{\partial t} = [\hat{H}(t), \hat{\rho}]$$

canonical ensemble

if $a=0$, $\hat{\rho}$ does not change and we assume $\hat{\rho}(0) = \frac{e^{-\beta \hat{H}_0}}{Z}$

Schwinger picture

$$\langle \hat{B}(t) \rangle = \text{Tr}[\hat{\rho}(t) \hat{B}]$$

change of $\hat{B}$ is purely due to the
 change of $\hat{\rho}(t)$ in eqn $\hat{B}$ is zero

$$\text{we assume } \langle \hat{B} \rangle_{H_0} = 0$$

interaction picture

$$\hat{B}_I(t) = e^{i \frac{H_0}{\hbar} t} \hat{B} e^{-i \frac{H_0}{\hbar} t}$$

$$\hat{\rho}_I(t) = e^{+i \frac{H_0}{\hbar} t} \hat{\rho}(t) e^{-i \frac{H_0}{\hbar} t}$$

$$\hat{\rho}_I(0) = \hat{\rho}(0)$$

$$i\hbar \frac{\partial \hat{\rho}_I}{\partial t} = e^{i \frac{H_0}{\hbar} t} [-H_0, \hat{\rho}] e^{-i \frac{H_0}{\hbar} t} + e^{i \frac{H_0}{\hbar} t} [\hat{H}_I, \hat{\rho}] e^{-i \frac{H_0}{\hbar} t}$$

$$= -a(t) \hat{A} = [\hat{V}_I, \hat{\rho}_I]$$

$$\text{Tr}[\hat{\rho}_I(t) \hat{B}_I(t)]$$

Let $\hat{\rho}_1(t) = S(t,0) \hat{\rho}_0 S(0,t)$ conjugate $\hat{\rho}_2 = e^{-\beta} \rho$
 where $S(t,0) = e^{\frac{iH_0 t}{\hbar}} U(t,0)$ Def and $\rho = U(\dots)$

$$i\hbar \frac{dS(t,0)}{dt} = \cancel{e^{\frac{iH_0 t}{\hbar}} H_0 e^{-\frac{iH_0 t}{\hbar}}} - H_0 e^{\frac{iH_0 t}{\hbar}} U(t,0) + e^{\frac{iH_0 t}{\hbar}} (\hat{H}_0 + \hat{V}) U(t,0) = \hat{V}_1(t) S(t,0)$$

integrate on both side

$$\int_0^t \frac{dS(t,0)}{dt} dt = \frac{1}{i\hbar} \int_0^t V_1(t') S(t',0) dt'$$

$$S(t,0) - 1 = \frac{1}{i\hbar} \int_0^t V_1(t') S(t',0) dt'$$

$$S(t,0) = 1 - \frac{i}{\hbar} \int_0^t V_1(t') S(t',0) dt'$$

iter

$$= 1 - \frac{i}{\hbar} \int_0^t V_1(t') dt' + \left(-\frac{i}{\hbar}\right) \int_0^t V_1(t') \int_0^{t'} V_1(t'') dt'' + \dots$$

Keep only to 1st order

$$\hat{\rho}_2(t) = \left[1 - \frac{i}{\hbar} \int_0^t V_1(t') dt' \right] \rho(0) \left[1 + \frac{i}{\hbar} \int_0^t V_1(t') dt' \right]$$

$$= \rho(0) - \frac{i}{\hbar} \int_0^t [V_1(t'), \rho(0)] dt' + O(V^2)$$

$$\langle \hat{B}(t) \rangle = \text{Tr} [\hat{\rho}_2(t) \hat{B}_2(t)]$$

$$V_1 = -\alpha(t) \hat{A}_1(t)$$

$$= \text{Tr} [\rho(0) \hat{B}_2(t)] - \frac{i}{\hbar} \int_0^t dt' \text{Tr} \{ [V_1(t'), \rho(0)] \hat{B}_2(t) \}$$

" $\hat{B}$ indep. is 0

$$\langle \hat{B}(t) \rangle = + \frac{i}{\hbar} \int_0^t dt' \text{Tr} \{ [A_1(t'), \rho(0)] \hat{B}_1(t) \} a(t')$$

$\xrightarrow{\quad}$
 $A_1 \rho B_1 - \rho A_1 B$
 $\rho B_1 A_1$

Date gy ch

$$= \frac{i}{\hbar} \int_0^t dt' \text{Tr} \{ \rho(0) [B_1(t), A_1(t')] \} a(t')$$

retarded Green's func

$$\text{Let } G_{BA}(t, t') = - \frac{i}{\hbar} \text{Tr} \{ \rho(0) [B_1(t), A_1(t')] \}$$

$$= - \frac{i}{\hbar} \langle [B_1(t), A_1(t')] \rangle_{eq} \theta(t - t')$$

$$\langle \hat{B}(t) \rangle = - \int_0^t dt' G(t, t') a(t')$$

general susceptibility in eq. $G_{BA}(t, t') = G_{AB}(t - t')$

$$\chi(t - t') = - G(t, t')$$

B current
A is drive
 $\dot{P} = j = B$

large time limit. in w. small
how current respond to electric
field gives conductivity

$$\hat{B}(\omega) = - G(\omega) a(\omega)$$

$d = \frac{B}{j}$
j is current
same conductivity

Fourier transform

$$\text{Tr} \{ \rho(0) [B_1(t), A_1(t')] \} = i \hbar \beta \langle \dot{B}_1(t); A_1(t') \rangle$$

$$\beta \langle \dot{a}; b \rangle = \frac{1}{i\hbar} \langle [a, b] \rangle$$

$$\langle \hat{B}(t) \rangle = - \beta \int_0^t dt' \langle \dot{B}_1(t); A_1(t') \rangle a(t')$$

$$\langle \dot{a}; b \rangle = \underbrace{- \langle a; \dot{b} \rangle}$$

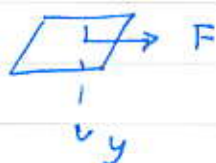
Kubo canonical correlation is

def as $\langle a; b \rangle = \frac{1}{\beta} \int_0^{\beta} d\lambda \text{Tr} [\rho_{(0)} e^{\lambda H_0} a e^{-\lambda H_0} b]$

Date

self
diffusion
constant

$$D = \int_0^{\infty} dt \langle v_x(t) v_x(0) \rangle$$

$$J_x = -x \nabla T$$


thermal
conductivity

$$\kappa = \frac{V}{k_B T^2} \int_0^{\infty} dt \langle J_x(t) J_x(0) \rangle$$

viscosity

$$\eta = \frac{V}{k_B T} \int_0^{\infty} dt \langle P_{xy}(t) P_{xy}(0) \rangle$$

$$\bar{A} = P_{xy} = -\eta \frac{\partial v_x}{\partial y}$$

we have $\langle a; b \rangle = \langle b; a \rangle$

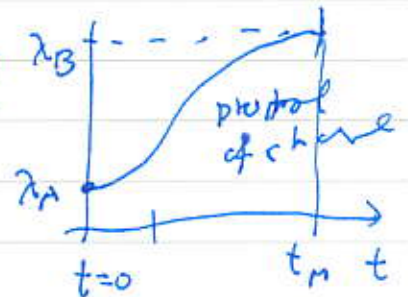
thermodynamics

$$H(\lambda)$$

fixing T means
the system is in contact with
a bath so that δQ is exchanged!

$$TdS \geq \delta Q = \delta W + dU$$

work done to the system infl. energy change



process that has fixed T

$$F = U - TS$$

$$dF = dU - TdS \quad \text{true only if } T \text{ is fixed}$$

the maximum
quasi-static
process

$$t_m \rightarrow \infty$$

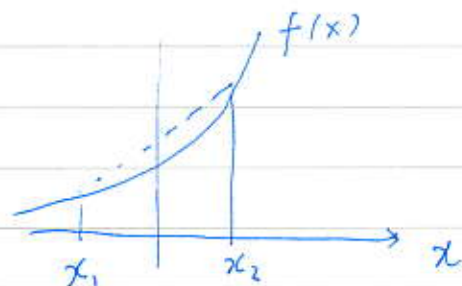
$$\delta W \geq dU - TdS = dF$$

integrate $W \geq F_B - F_A = \Delta F \quad = \text{if } T_m \rightarrow \infty$

Jarzynski equality [PRL 78, 2690 (1997)]

$$\overline{e^{-\beta W}} \leq \overline{e^{-\beta F}} = e^{-\beta \Delta F} \quad \text{for finite } T.$$

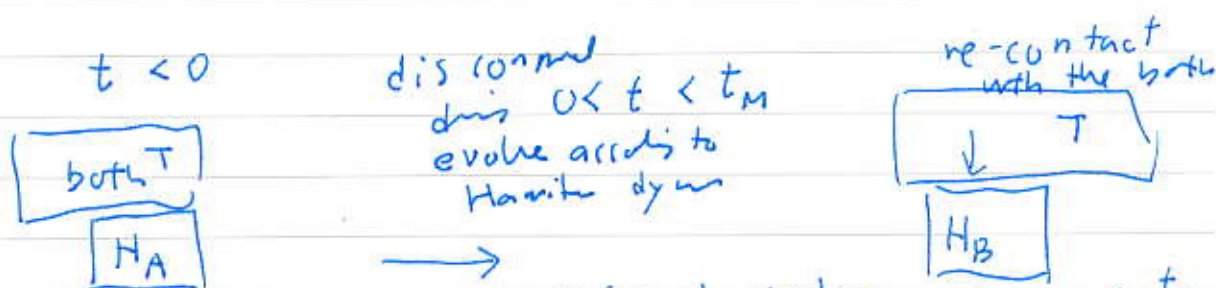
$$\overline{e^x} \geq e^{\overline{x}} \quad \text{exponential function is convex}$$



$$\lambda f(x_1) + (1-\lambda) f(x_2) \geq f(\lambda x_1 + (1-\lambda) x_2)$$

$$\langle f \rangle \geq f(\langle x \rangle)$$

Keep the heat-bath in the picture (Nosé-Hoover or coupled bigger system)
this is more difficult.



Work to the system

$$W = \int_0^{t_m} dt \lambda \frac{\partial H}{\partial \lambda}$$

~~the~~ $H(\lambda(t))$ system out of equilibrium at t_m but can be reestablished during contact with bath again

$$Z = \left(\frac{P}{q} \right)$$

depends on the path of the dynamics in phase space

$z(t)$ is deterministic fixed

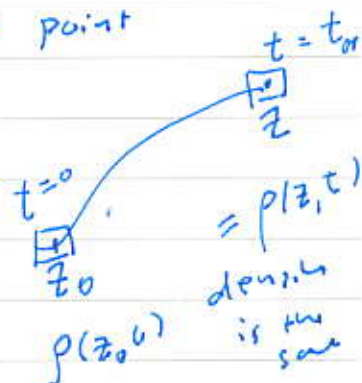
only by the initial condition $z(0)$

" — " avg means avg over initial conditions expressed by the begin point

$$\overline{e^{-\beta W}} = \int dz_0 p(z_0, 0) e^{-\beta W(z_0, t_m)}$$

$$= \int dz p(z, t) e^{-\beta W(z, t)}$$

distribution at time t



$$W = \int_0^t dt \lambda \frac{\partial H}{\partial \lambda} = \int d\lambda \frac{\partial H}{\partial \lambda} = H(\lambda, z) - H(\lambda_0, z_0)$$

$dz_0 = dz$ become hamilton

dynamics preserve phase space volume

and ρ satisfies Liouville then $\frac{d\rho}{dt} = \{H, \rho\}$ or $\frac{D\rho}{Dt} = 0$

begin point

end point

$$\overline{e^{-\beta W}} = \int dz \, p(z, t) e^{-\beta (H(\lambda_B, z) - H(\lambda_A, z_0))}$$

↓ back to $t=0$

$$= \int dz \, \underbrace{p(z_0, 0) e^{+\beta H(\lambda_0, z_0)}}_{\text{cancel}} e^{-\beta H(\lambda_B, z)}$$

only thing we
really need is
 $\int p(z_0, 0) dz_0 = 1$

$$= \frac{1}{Z_A} \int dz \, e^{-\beta H(\lambda_B, z)} = \frac{Z_B}{Z_A} = \frac{e^{-\beta F_B}}{e^{-\beta F_A}} = e^{-\beta (F_B - F_A)} = e^{-\beta \Delta F}$$

$F = -k_B T \ln Z$

Gallavotti-Cohen Relation (theorem)



$$\frac{P(Q)}{P(-Q)} = e^{Q \left(\frac{1}{k_B T_R} - \frac{1}{k_B T_L} \right)}$$

$$= e^{Q (\beta_R - \beta_L)}$$

$P(Q)$ what is
the heat
transfer from left
to right lead
during a fixed
time t ?

$$\beta = \frac{1}{k_B T}$$

13 Jan 2014

0-th law of thermodynamics

Date

If, of three bodies, A, B and C, A and B are separately in equilibrium with C, then A and B are in equilibrium with one another.

→ empirical temperature $\theta = \phi(P, V)$ can be defined
Bodies in equilibrium have the same temperature

1st law

Energy is conserved if heat is taken into account.

U is a state function.

$$\Delta U = W + Q$$

Clausius

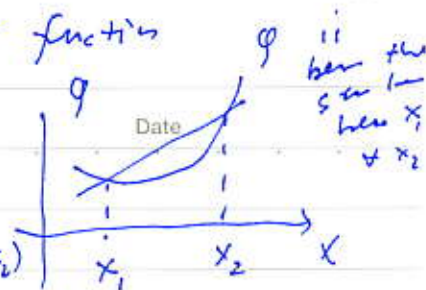
2nd law

It is impossible to devise an engine which, working in a cycle, shall produce no effect other than the transfer of heat from a colder to a hotter body.

Kelvin

It is impossible to devise an engine which, working in a cycle, shall produce no effect other than the extraction of heat from a reservoir and the performance of an equal amount of mechanical work.

Jensen's inequality if $\varphi(x)$ is a convex function
then $\overline{\varphi(x)} \geq \varphi(\bar{x})$



$$\varphi(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 \varphi(x_1) + \lambda_2 \varphi(x_2)$$

$$0 \leq \lambda_1, \lambda_2 \leq 1 \\ \lambda_1 + \lambda_2 = 1$$

straight line
parameter by λ

$e^{\pm x}$ is convex

$$H = H_0 + H'$$

$$Z = \text{Tr}(e^{-\beta H})$$

$$Z_0 = \text{Tr}(e^{-\beta H_0})$$

$$e^{-\beta H} = e^{-\beta H_0} e^{-\beta H'}$$

Jensen inequality

$$\langle \dots \rangle_0 = \frac{\text{Tr}(e^{-\beta H_0} \dots)}{Z_0}$$

$$\frac{Z}{Z_0} = \langle e^{-\beta H'} \rangle_0 \geq e^{-\beta \langle H' \rangle_0}$$

$$\ln Z - \ln Z_0 \geq -\beta \langle H' \rangle_0$$

divided by $-\beta$

$$F - F_0 \leq \langle H' \rangle_0 = \langle H - H_0 \rangle_0$$

$$F \leq F_0 + \langle (H - H_0) \rangle_0$$

Feynman - Jensen inequality
or Bogoliubov inequality

Ising model

$$H_0 = -h_{\text{eff}} \sum_{i=1}^N \sigma_i$$

$$[h_{\text{eff}} = J g m + h]$$

$$F_0 = -k_B T N \ln Z = -k_B T N \ln 2 \cosh(\beta h_{\text{eff}})$$

deter h_{eff} by minimization

$$\langle H - H_0 \rangle_0 = \langle -J \sum_{i,j} \sigma_i \sigma_j - (h - h_{\text{eff}}) \sum_{i=1}^N \sigma_i \rangle_0$$

$$= -J \frac{g}{2} N m^2 - (h - h_{\text{eff}}) m \cdot N$$

$$m \equiv \langle \sigma_i \rangle_0$$

$$F \leq -k_B T N \ln(2 \cosh(\beta h_{\text{eff}})) + \frac{1}{2} J q N m^2 - (h - h_{\text{eff}}) \cdot m \cdot N$$

min w.r.t to h_{eff} & m

$$1) \quad \frac{\partial \psi}{\partial h_{\text{eff}}} = 0 \quad \text{but} \quad -\frac{N}{\beta} \frac{\sinh(\beta h_{\text{eff}})}{2 \cosh(\beta h_{\text{eff}})} \cdot \beta + m N = 0$$

$$\rightarrow m = \tanh(\beta h_{\text{eff}})$$

$$2) \quad 0 = \frac{\partial \psi}{\partial m} = -J q N m - (h - h_{\text{eff}}) N = 0 \quad h_{\text{eff}} = h + J q m$$

approx free energy is

$$F \approx -k_B T N \ln(2 \cosh(\beta(h + J q m))) + \frac{1}{2} J q N m^2$$

Or Vilk's method

A. Georges "Dynamical mean-field theory"

RMP 68 13 (1996)

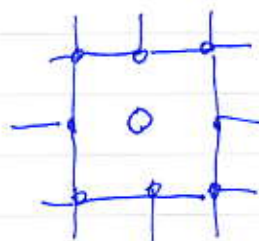
$q = 2D$ hypercubic lattice

$$k_B T_c = J q$$

$$\text{side } J \rightarrow \frac{J}{2D} = J^*$$

$$J \rightarrow 0$$

$$\frac{k_B T_c}{J^*} = 1$$



$$H = -h_0 \sigma_0 - J \sum_{j \in \text{nn of } 0} \sigma_j \cdot \sigma_0 + H_0$$

↑
site 0
as well into
next 0 & nn
removed

effective Hamiltonian

$$e^{-\beta H_{\text{eff}}(\sigma_0)} = \sum_{\text{sum over all except site } \sigma_0} e^{-\beta H(\sigma)}$$

Exact stat mech & cavity

$$Z_0 = \sum_{\sigma_i \quad i \neq 0} e^{-\beta(-\sum \eta_i \sigma_i + H_0)}$$

$$\langle \sigma_i \sigma_j \dots \sigma_k \rangle_c^{(0)} = \frac{\partial^n \ln Z_0}{\partial \eta_i \partial \eta_j \dots \partial \eta_k} \bigg|_{\eta_i = 0}$$

Taylor expand $\ln Z_0$

$$\ln Z_0(\eta) = \ln Z_0(\eta=0) + \beta \sum_i \langle \sigma_i \rangle^{(0)} \eta_i + \frac{1}{2} \beta^2 \sum_{i,j} \langle \sigma_i \sigma_j \rangle_c^0 \eta_i \eta_j + \dots$$

↓
→ 0 $D \rightarrow \infty$

$$e^{-\beta H_{\text{eff}}(\sigma_0)} = \sum_{\{\sigma_i\}} e^{-\beta H(\sigma)}$$

$$= \sum_{\{\sigma_i\}} e^{-\beta(-h\sigma_0 - \frac{J^*}{2D} \sum_j \sigma_j \sigma_0 + H^0)}$$

special case of spin cavity hamiltonian

$$= e^{\beta h \sigma_0} \sum_{\{\sigma\}} e^{-\beta(\sum_j \sigma_j \eta_j + H^0)} \quad \eta_j = \begin{cases} \frac{J^*}{2D} \sigma_0 & \text{if } j \in \text{nn } \sigma_0 \\ 0 & \text{otherwise} \end{cases}$$

take $\ln$ due by $-\frac{1}{\beta}$

$$H_{\text{eff}}(\sigma_0) = -h\sigma_0 - \frac{1}{\beta} (\ln Z_0(\eta) + \dots)$$

$$= \text{const} - h\sigma_0 - \sum_{j \in \text{nn } \sigma_0} \langle \sigma_j \rangle^0 \eta_j + \dots$$

$$\text{as } D \rightarrow \infty \quad \langle \sigma_j \rangle^0 = \langle \sigma_j \rangle + O\left(\frac{1}{D}\right)$$

$$H_{\text{eff}}(\sigma_0) = -h\sigma_0 - \cancel{\sum} \langle \sigma_0 \rangle J \cdot q \sigma_0 + \dots$$